

CSC 2400-002: Design of Algorithms (Spring 2026)

Tennessee Tech University

Drill 6 Solutions

Topic: Key Properties and Applications of Asymptotic Notation

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Question 1. Let B be the number of bits in the binary representation of a decimal number N .

Select all asymptotic notation that you can correctly put in place of ? in the equation below.

$$B = ?(\log_{10} N)$$

- ☐ O (Big-Oh)
- ☐ Ω (Big-Omega)
- ☐ Θ (Big-Theta)

Solution: From class, we know that $B = \lfloor \log_2 N \rfloor + 1 \in \Theta(\log N)$ where we can put any constant base on the log, and in particular base 10. Hence, $B \in \Theta(\log_{10} N)$, and so it is in $O(\log_{10} N)$ and $\Omega(\log_{10} N)$ as well.

Question 2. Suppose that Algorithm 1 has runtime $\Theta(n^3)$ and Algorithm 2 has runtime $\Theta(n^2 \log n)$. Algorithm 3 is one that runs Algorithm 1 followed by Algorithm 2. Select all sets below where the runtime of Algorithm 3 belongs.

- ☐ $\Theta(n^3)$
- ☐ $O(n^5)$
- ☐ $\Omega(n \log n)$
- ☐ $\Theta(n^2 \log n)$

Solution:

Let $T_1(n)$ and $T_2(n)$ be the runtimes of Algorithm 1 and Algorithm 2 respectively. Since Algorithm 3 runs them one by one, its runtime is $T_1(n) + T_2(n)$. Since, $T_1(n) \in \Theta(n^3)$ and $T_1(n) \in \Theta(n^2 \log n)$, by the analogs of Theorems 4-5 of lec6.pdf for $\Theta()$, we have $T_1(n) + T_2(n) \in \Theta(n^3 + n^2 \log n) = \Theta(n^3)$ since $n^3 + n^2 \log n \in \Theta(n^3)$.

Now, any function in $\Theta(n^3)$ must be $\leq c \cdot n^3 \leq c \cdot n^5$ for some constant c and large enough n ; hence, the runtime is also in $O(n^5)$.

Again, any function in $\Theta(n^3)$ must be $\geq c \cdot n^3 \geq c \cdot n \log n$ for some constant c and large

enough n ; hence, the runtime is also in $\Omega(n \log n)$.

But a function in $\Theta(n^3)$ must be $\geq c \cdot n^2 \log n$ for any c for all large enough n . So the runtime is not in $O(n^2 \log n)$, and hence, not in $\Theta(n^2 \log n)$.

Question 3. Formally prove (either using definitions or the limits theorem):

For any two functions $f(n)$ and $g(n)$, we have $f(n) + g(n) \in \Theta(\max(f(n), g(n)))$

Solution: Let $h(n) := \max(f(n), g(n))$. Then, since each of $f(n), g(n) \leq h(n)$, we have $f(n), g(n) \in O(h(n))$, which means $f(n) + g(n) \in O(h(n))$ by Corollary 2 in lec6.pdf.

(Without explicitly using Corollary 2, you can also show it by basically using its proof:

$f(n) + g(n) \leq h(n) + h(n) = 2 \cdot h(n)$ for any $n \geq 1$, which means $f(n) + g(n) = O(h(n))$.)

Again, $f(n) + g(n) \geq h(n)$ for all $n \geq 1$, since $h(n)$ is either $f(n)$ or $g(n)$ for any n . Therefore, $f(n) + g(n) \in \Omega(h(n))$. Thus, we conclude $f(n) + g(n) \in \Theta(h(n)) = \Theta(\max(f(n), g(n)))$.

Alternate solution using limits: Let $h(n) := \max(f(n), g(n))$. Then,

$$\lim_{n \rightarrow \infty} \frac{f(n) + g(n)}{h(n)} = \lim_{n \rightarrow \infty} \left(\frac{f(n)}{h(n)} + \frac{g(n)}{h(n)} \right) \leq 1 + 1 = 2 < \infty$$

since each of $f(n), g(n) \leq h(n)$.

Again, since $f(n) + g(n) \geq h(n)$ for all n as argued above, we have

$$\lim_{n \rightarrow \infty} \frac{f(n) + g(n)}{h(n)} \geq 1 > 0$$

Therefore, by the limits theorem, $f(n) + g(n) \in \Theta(h(n)) = \Theta(\max(f(n), g(n)))$